

CP2 Project Logbook

Semester 2 Weekly Progress Record | Week 1-12 | 11 May-31 July 2026

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Project title: Evidence-Grounded Local-LLM Explanations for Credit Card Fraud Alert Review

Semester period: Official CP2 Week 1 to Week 12, 11 May to 31 July 2026

This logbook is organised according to the official Capstone Project 2 teaching weeks. The early entries record the transition from the approved CP1 proposal into an implementable CP2 design. Later entries reference the dated code, experiments, reports, and evaluation artifacts produced during implementation. The evidence column identifies the project material associated with each work stream. No supervisor meeting, working hour, or feedback is claimed unless it formed part of the actual project process. The first entry therefore starts at the beginning of the CP2 semester timeline, not only at the July implementation period.

Weeks 1-9 are retrospective CP1-to-CP2 planning and design summaries reconstructed against the official teaching-week timeline. Dated repository implementation and experiment evidence begins in Week 10; later-created files listed in the early entries show where each design was subsequently formalised.

Week 1. Project Restart and CP1-to-CP2 Scope Review

Date: 11-17 May 2026

Weekly objective: Restart the project from the approved CP1 proposal and identify what had to be delivered during CP2.

Progress and output: I reviewed the CP1 proposal, including the original Autoencoder-XGBoost detector, SHAP explanations, local LLM, privacy rationale, and guardrail plan. I separated the work into two questions: whether the hybrid detector improved fraud detection, and whether generated explanations could be kept faithful to model evidence. This prevented the LLM from being treated as a second fraud detector.

Evidence/files: CP1/01_FINAL_SUBMISSION/Proposal_Capstone_Project.pdf; final report Sections 1.2-1.5 and 3.1.

Problem or decision: The proposal combined model development, explainable AI, an LLM, and an application. Without a clear hierarchy, CP2 could have looked like several unrelated projects.

Critical reflection: I decided that the detector would remain the source of the prediction, SHAP would provide attribution evidence, and the LLM would only articulate that evidence. This role separation later became the central project design.

Next-week action: Confirm project continuity and map the semester deliverables.

Week 2. Supervisor Continuity and Semester Planning

Date: 18-24 May 2026

Weekly objective: Confirm the CP2 administrative starting point and plan the work against the Week 12 submission deadline.

Progress and output: I confirmed that Dr Claymond Lim Wei Xiang remained the supervisor from CP1. I mapped the project work to the CP2 teaching timeline: design and preparation first, implementation and evaluation next, followed by the final report, logbook, presentation, and system demonstration. I also kept the project title open to revision if the detector results did not support the original Autoencoder emphasis.

Evidence/files: docs/references/CP2_Timeline_2026.png; CP1 proposal cover; final report cover and Section 1.3.

Problem or decision: A schedule based only on model training would leave insufficient time for explanation validation, application work, writing, and demonstration preparation.

Critical reflection: Treating the report, reproducibility evidence, and system demonstration as project outputs from the beginning reduced the risk of building a model that could not later be explained or defended.

Next-week action: Revisit the literature and identify the most defensible research gap.

Week 3. Literature Review and Contribution Reframing

Date: 25-31 May 2026

Weekly objective: Reassess which part of the proposed project could make a credible FYP contribution.

Progress and output: I reviewed literature on imbalanced fraud detection, SMOTE, cost-sensitive learning, autoencoder features, XGBoost, SHAP, narrative explanations, local language models, hallucination, and programmable guardrails. Existing work already covered hybrid fraud detectors and LLM-based explanation. The more specific gap was the measurement and fail-closed control of LLM text against fixed SHAP evidence.

Evidence/files: The starting literature foundation is recorded in the CP1 literature review. The resulting CP2 synthesis was later formalised in final report Chapter 2 and Table 2.1 and verified in docs/reviews/2026-07-26-final-reference-audit.md.

Problem or decision: Claiming that Autoencoder plus XGBoost or “using an LLM for explanations” was new would be difficult to defend against prior research.

Critical reflection: I reframed the intended contribution as an evaluated explanation-delivery boundary: local generation, minimized evidence, code-level validation, measured failure rates, and deterministic fallback. The detector benchmark would test the original CP1 assumption rather than guarantee it.

Next-week action: Define the detector comparison, metrics, and research questions before implementation.

Week 4. Experimental Groups and Evaluation Design

Date: 1-7 June 2026

Weekly objective: Design a fair detector experiment that could test the CP1 hybrid-model assumption.

Progress and output: I specified six detector groups: baseline XGBoost, SMOTE-XGBoost, reconstruction-error hybrid, reconstruction-error plus SMOTE, cost-sensitive XGBoost, and latent-feature hybrid. Five fixed seeds were planned for every group. Average Precision was selected as the primary ranking metric, with precision, recall, F1, ROC-AUC, ranked-alert metrics, confusion counts, and runtime retained for interpretation.

Evidence/files: This design was later formalised in final report Sections 3.4-3.5 and Tables 3.1-3.2 and implemented in configs/g0_xgb.yaml to configs/g7_ae_latent_xgb.yaml.

Problem or decision: Fraud accuracy is misleading because non-fraud cases dominate the dataset. A single metric could also hide operational trade-offs between missed frauds and unnecessary reviews.

Critical reflection: The multi-group, multi-seed design allowed the original hybrid idea to fail without invalidating the project. It also created a stronger basis for choosing one frozen detector for later SHAP and narrative work.

Next-week action: Define the data split, preprocessing, identity, and leakage controls.

Week 5. Dataset and Leakage-Control Design

Date: 8-14 June 2026

Weekly objective: Establish a reproducible data protocol before model training.

Progress and output: I retained the European Credit Card Fraud dataset from the CP1 proposal and defined content deduplication, stable case identifiers, a stratified 70/15/15 train-validation-test split, train-only scaler fitting, and train-only SMOTE. Validation data would be used for tuning, detector selection, and threshold selection; test data would remain untouched until freezing.

Evidence/files: This protocol was later formalised in `data/raw/DATA_MANIFEST.md` and final report Sections 3.2-3.3, then implemented in `src/data/load.py`, `src/data/split.py`, `src/data/preprocess.py`, and `src/data/resample.py`.

Problem or decision: Leakage could occur through duplicate transactions, scaling, resampling, autoencoder fitting, threshold selection, or unsafe joins between predictions and explanations.

Critical reflection: Stable `case_id` values became necessary because SHAP, LLM, and application records should never be joined by row position. I also recognised that leakage prevention had to be demonstrated by tests and manifests, not only described in the report.

Next-week action: Finalise the Autoencoder and XGBoost component design.

Week 6. Detector Architecture and Hybrid-Feature Plan

Date: 15-21 June 2026

Weekly objective: Translate the model proposal into components that could be compared under one interface.

Progress and output: I planned a common XGBoost wrapper, validation-only early stopping, validation-selected thresholding, an Autoencoder trained on legitimate training observations, reconstruction-error features, and latent features. SMOTE and class weighting were kept as separate imbalance strategies so their effects would not be mixed.

Evidence/files: The component design was later formalised in final report Sections 3.4-3.6 and implemented in `src/models/xgb.py`, `src/models/autoencoder.py`, `src/evaluation/metrics.py`, and `src/evaluation/threshold.py`.

Problem or decision: The original title could encourage selecting the hybrid model regardless of the result. I instead planned all groups to use the same split, metrics, artifact contract, and seeds.

Critical reflection: A negative Autoencoder result would still answer a valid research question. This was more academically useful than changing the metric or comparison after seeing the outcome.

Next-week action: Define the SHAP-to-LLM evidence boundary and narrative failure checks.

Week 7. Explainability, Local LLM, and Guardrail Design

Date: 22-28 June 2026

Weekly objective: Specify how generated text would remain subordinate to detector evidence.

Progress and output: I designed signed SHAP reason codes containing feature identity, rank, and contribution direction. The local Ollama model would receive only a minimized reason-code package, not a raw transaction row, exact values, labels, probabilities, or SHAP magnitudes. Candidate narratives would be checked for format, completeness, grounding, and direction. A failed check or unavailable LLM would return a deterministic brief.

Evidence/files: This boundary was later formalised in final report Sections 3.6-3.8 and implemented in `src/explainability/`, `src/narratives/`, and `corpus/guardrail_corpus_v1.jsonl`.

Problem or decision: Automatically normalizing or repairing rejected LLM text would make the output look correct while hiding what the model originally generated.

Critical reflection: I separated OFF-policy measurement of raw output from ON-policy delivery. This allowed the project to report the LLM's observed failure rate while ensuring that detected failures were not shown as accepted evidence.

Next-week action: Design an application that demonstrates the same evaluated evidence chain.

Week 8. Analyst Workbench Architecture

Date: 29 June-5 July 2026

Weekly objective: Plan a deliverable application that resembles an analyst workflow rather than a model demonstration page.

Progress and output: I selected React and TypeScript for the frontend and FastAPI for the local backend. The planned workflow used one alert queue, a case review workspace, explanation assurance, evaluation results, and separate analyst actions. Research artifacts would be immutable consumers, while status, notes, and activity would be stored in a separate local SQLite workflow plane.

Evidence/files: This architecture was later formalised in docs/specs/2026-07-13-react-fastapi-demo-dashboard-spec.md and final report Sections 3.10-3.11, then implemented in app/frontend/ and app/backend/.

Problem or decision: A dashboard organised around tabs for models and charts could appear fragmented and would not explain who uses the system or what action follows an alert.

Critical reflection: The application was reframed around the sequence “select alert, inspect evidence, review validation, record an action.” It would support decision-making but would not declare whether a transaction was truly fraudulent.

Next-week action: Prepare the implementation contracts, provenance schema, and review criteria.

Week 9. Implementation Readiness and Evidence Contracts

Date: 6-12 July 2026

Weekly objective: Convert the design into an executable build sequence with clear artifact boundaries.

Progress and output: I prepared the implementation work breakdown for data, models, experiments, SHAP, narrative generation, provenance, results, and the application. I defined run manifests, stable source-run links, frozen artifacts, and adversarial review expectations. Tests were planned where they protected leakage, provenance, validation, and workflow contracts, rather than forcing every documentation or presentation task into TDD.

Evidence/files: This readiness work was formalised at the start of Week 10 in docs/plans/2026-07-13-cp2-implementation-plan.md, docs/specs/2026-07-13-react-fastapi-demo-dashboard-spec.md, and docs/specs/2026-07-14-fraud-review-workbench-spec.md.

Problem or decision: The system had many interfaces. If detector predictions, SHAP records, narratives, and dashboard cases used inconsistent identities or unverified “latest” directories, the final demonstration could silently show a different system from the reported experiment.

Critical reflection: Treating the plan and manifests as contracts made later results easier to audit. The most important tests were placed at research-risk boundaries instead of using TDD mechanically for every task.

Next-week action: Implement and execute the full detector-to-explanation pipeline.

Week 10. Core Implementation, Detector Benchmark, and First Review Package

Date: 13-19 July 2026

Weekly objective: Build the reproducible research pipeline and obtain the first complete experimental evidence.

Progress and output: I implemented dataset integrity checks, stable case IDs, preprocessing, SMOTE, the XGBoost and Autoencoder components, thresholding, manifests, and leakage audits. I ran six groups over five seeds, producing 30 detector runs. G6 had the numerically highest mean test AP at 0.855214, although the leading differences were small and the Autoencoder variants showed no clear advantage. G6 seed 42 was frozen for SHAP. I generated reason codes for 51 flagged cases, ran the strict and simple Ollama prompt arms, calibrated the validator, implemented the first React-FastAPI workbench, and prepared a supervisor review package.

Evidence/files: Git commits from dc19eb8 onward; experiments/runs/; reports/tables/results_summary.csv; experiments/runs/2026-07-14_g4_seed42/; experiments/runs/2026-07-14_g5_seed42/; app/; supervisor_meeting/.

Problem or decision: The original Autoencoder claim was not supported. The ULB feature names were also anonymous, so LLM narratives remained close to the SHAP reason-code wording.

Critical reflection: I kept the negative detector result and repositioned the project around verifiable explanation delivery. The first workbench made the evidence visible but still needed a clearer operational story and better result visualisation.

Next-week action: Present the progress, respond to feedback, and strengthen the readable explanation evaluation.

Week 11. Supervisor Feedback, Result Visualisation, and S0 Evaluation

Date: 20-26 July 2026

Weekly objective: Improve how the project communicates detector trade-offs and demonstrate the value and limitations of the local LLM more clearly.

Progress and output: During the supervisor discussion, the main practical feedback was to show more bar charts, especially F1, and to make the purpose of the website and the value of Ollama clearer. I added mean F1 with standard-deviation error bars, grouped precision and recall, false-positive and false-negative counts, and narrative violation, fallback, and acceptance charts. I also ran an exploratory model search but retained the frozen G6 chain because the alternatives did not justify breaking provenance. To address the anonymous ULB limitation, I added the S0 synthetic stream with readable transaction concepts. Its frozen test result was AP 0.544017 and F1 0.514286. Among 25 selected alerts, 23 guarded LLM briefs passed and 2 used fallback.

Evidence/files: docs/reviews/2026-07-20-detector-model-search.md;
reports/figures/detector_metric_bars.png; reports/figures/narrative_delivery_bars.png;
docs/specs/2026-07-26-semantic-fraud-triage-extension.md; experiments/runs/2026-07-26_s0_seed42/; reports/semantic_results_manifest.json.

Problem or decision: S0 could make the project appear to contain two unrelated detectors. The local LLM also did not consistently produce a more complete brief than the deterministic renderer.

Critical reflection: I defined ULB as the supporting real-data detector benchmark and S0 as the primary readable semantic and operational context. The LLM's value is controlled articulation and measurable failure handling, not new risk knowledge. The accepted S0 outputs choosing shorter summaries was retained as a negative result.

Next-week action: Complete the human pilot, final report, presentation, and submission verification.

Week 12. Human Pilot, Final Report, Presentation, and Submission Preparation

Date: 27-31 July 2026

Weekly objective: Integrate the completed evidence into the assessed CP2 deliverables before the 31 July deadline.

Progress and output: I prepared the consent material, questionnaire, recruitment message, analysis plan, and explanation stimuli. Eleven adult proxy reviewers completed 99 case reviews. Because this was below the planned minimum of 18 and target of 30, I reported it as descriptive interim evidence and did not run a superiority test. I also manually audited all 49 strict-arm narratives delivered without fallback against their bound evidence and found no semantic violation; the 95% Wilson interval was 0%-7.27%, so I retained the single-reviewer and uncertainty limits. I completed the rubric-aligned final report, 12-week logbook, 12-slide presentation, presenter guide, submission index, rubric matrix, page-map verification, and software test record. As of 29 July, the final report was 79 A4 pages and the tested application remained a local prototype rather than a production-bank deployment.

Evidence/files: docs/studies/; reports/human_eval_results.json; reports/tables/human_eval_*.csv; reports/thesis/submission/; reports/presentation/; reports/RUBRIC_ALIGNMENT_MATRIX.md; reports/thesis/submission/VERIFICATION_RECORD.md; experiments/audit/audit_result.json.

Problem or decision: The human sample was smaller than planned, and the submission still required consistency checks across the report, logbook, slides, figures, page numbers, and stated limitations.

Critical reflection: The final package is strongest when it reports both positive and negative evidence. The project demonstrates a reproducible detector-to-explanation chain and a fail-closed local-LLM delivery boundary, but the 0/49 manual-audit observation still does not prove universal semantic correctness. It also does not prove a new best detector, improved analyst productivity, or production readiness.

Next action: Submit the final report and logbook by the course deadline, rehearse the system demonstration and viva, and continue recruitment only if additional responses can be incorporated without changing the frozen evidence claims improperly.

Overall Critical Reflection

Across the 12 teaching weeks, the project moved from the CP1 hybrid-model proposal to an evidence-grounded explanation-delivery study. The Autoencoder variants were implemented and

evaluated but did not show a clear advantage. That result changed the emphasis without being removed from the report. The local LLM was then constrained to a narrow role: translating minimized SHAP evidence into a standardised brief. Its raw output was measured separately from the fail-closed delivery policy, and detected failures returned deterministic evidence.

The application, S0 study, and interim human pilot improved the practical and semantic evaluation, but they also introduced boundaries that must remain visible. S0 is synthetic, the ULB and S0 detector scores are not directly comparable, the validator only covers its tested contract, and the human pilot contains 11 proxy reviewers rather than the planned 30. Preserving these limits made the final project more credible and easier to defend during the viva.